19	In a spectrometer experiment, light of wavelength 400 nm is incident normally on a diffraction grating having 400 lines per millimeters. What is the angle of diffraction of the third order diffracted beam?						
	A	13.9°	**B**	18.7°	**C**	28.7°	**D**	56.1°

Answer: C

$d\sin\theta = n\lambda$

$\sin\theta = n\lambda/d$ $\qquad$ $1/d = 400 \times 10^3$

$\sin\theta = 3(400\text{ nm})(400 \times 10^3) \rightarrow \theta = 28.7°$

20	Three point charges, each of magnitude Q, are placed at the three corners of a square as